



FIRST SEMESTER 2026-2027
PHA G613 Pharmaceutical Biotechnology
Course Handout (Part-II)

Date: 01/08/2026

In addition to Part-I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No : **PHA G613**
Course Title : Pharmaceutical Biotechnology
Instructor-in-charge : Dr. Gautam Kumar
Instructors : To be announced by the department

Scope and objective of the course:

The course deals with the principles and techniques of biotechnology with direct application in the field of pharmaceutical sciences. It aims at bringing together the principles of molecular biology, molecular genetics, biochemical engineering, protein engineering, enzymotechnology, bio-informatics required to develop a successful biotechnology based pharmaceutical product. The applications of genetic engineering, recombinant DNA technology, hybridoma technology, transgenic animals, gene therapy, drug delivery aspects of biotechnology products and their clinical evaluation are also dealt with in this course.

1. Text Book (TB):

Bernard R. Glick, Jack J. Pasternak, Molecular Biotechnology- Principles and Application of rDNA. 2nd edition, ASM Press, Washington, 1998.

2. Reference Books (R):

- R1:** Brown T. A., Gene Cloning & DNA Analysis An Introduction 6th edition. Blackwell Publishing, UK, 2010.
R2: Balasubramanian D., Bryce, C.F.A., Dharmalingam, K., Green, J., Jayaraman, K. Concepts in Biotechnology, Urvee University Press, Hyderabad, India, 1996.





3. Course Plan:

Module Number	Learning objective	Reference	Learning Outcome
1. Introducing to pharmaceutical biotechnology	L.1. Introducing to pharmaceutical biotechnology	TB: Chap 1	Exposure to cell biology and pharmaceutical biotechnology.
2. Theoretical basis of molecular biotechnology	L.2-L.4. DNA replication L.5-L.6. DNA Transcription L.7-L.8. Protein synthesis (To bring students from different background to the same level)	Class notes and slides	Knowledge of structural & functional dynamics of cell; Central dogma; Biological systems used in biotechnology. Theoretical understanding of chemical synthesis and in vitro experiments using DNA
	L.9-L.11. Chemical synthesis of DNA	TB: Chap 5	
3. Recombinant DNA technology	L.12-L.18.. Recombinant DNA technique	TB: 4 R1: Chap 4, 5, 6, 7	Intensive understanding of r-DNA technology. Detailing of the challenges and solutions of gene expression using suitable vectors
	L.19-L.24. Application of r-DNA technology in synthesis of commercial products	TB: Chap 6, 7, 10	
4. Genetic engineering	L. 24-L.27. Application of mutagenesis in protein engineering	TB: Chap 8	Application of mutagenesis in protein engineering. Techniques to conduct genetic engineering in animals and plants.
	L. 28-L.31. Genetic engineering in animals and their application	TB: Chap 19	
	L.32-L.33. Genetic engineering in plants and their application	TB: Chap 17, 18	
5. Immuno-biotechnology	L.34-L.36. Application of biotechnology & immunotechnology in molecular diagnostics	TB: Chap 9	Knowledge of the use of biotechnology tools in diagnosis of diseases.
6. Industrial biotechnology	L.37-L.38. Principles of biochemical engineering in the production of biotechnology based products	TB: Chap 16	Exposure to commercial production of biotechnological products.
7. Application of biotechnology in human therapy	L.39-L.40. Understanding bio-informatics; high throughput DNA sequencing and proteomics	Class notes and journal references	Students will acquire basics information of bioinformatics and high throughput technique. Exposure to the application of biotechnology in human therapy is expected to gain.
	L.41. Gene cloning and DNA analysis in medicine	Class noted and Journal articles	





4. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Remarks
Mid-Semester Test	90 min.	25		Closed Book
Laboratory Components	-	20	Throughout the semester	Closed /Open Book
Assignments / Seminar/quiz	-	15	Throughout the semester	Closed/ Open Book
Comprehensive Exam	3 hrs.	40		Closed

- 5. Mid-Semester Evaluation:** Will be announced after Mid Sem Test.
- 6. Chamber Consultation Hours:** To be announced in the class.
- 7. Make-up:** Prior approval or intimation to take a make-up is a must. It is solely the discretion of the instructor-in-charge dependent upon the genuineness of the circumstances to allow a student to appear for a make-up evaluation component.
- 8. Grading policy:** As specified in Handout – Part I, appended to the timetable. The instructor in-charge reserves the right to award a NC report in case the student does not make himself or herself available for any of the evaluation component mentioned above. Also it is not imperative on part of the instructor in charge to award all the grades. Borderline cases during grading will be judged on the basis of regularity to classes and consistency or progress in the performance in evaluation components. The maximum pull-up to be exercised by the instructor in-charge will be announced in the class and shall be based on the subjective judgment of the evaluator.

Instructor-in-charge
PHA G613

